

JACK NAIMER

97 Ava Road | Toronto ON | M6C1V9 | +1 (647) 926-9723 | jack.naimer@mail.utoronto.ca

EDUCATION & TRAINING

Engineering Science – Machine Intelligence Major, University of Toronto 2021-2025

- Rigorous program famous for its intense academic challenge.
- Wide range of machine learning specialty courses and advanced courses in mathematics, physics, biology, and chemistry.
- Dean's Honours List – Fall 2024, Winter 2024, Fall 2023, Fall 2021

Math and Computer Science (MaCS) enriched high school program, William Lyon Mackenzie Collegiate Institute 2017-2021

- Achieved 5/5 on the college-level AP Computer Science exam in 10th grade.

PUBLICATIONS

[Preprint \(arXiv:2410.15185\): “Semantically Safe Robot Manipulation: From Semantic Scene Understanding to Motion Safeguards”](#) – Accepted to *IEEE Robotics and Automation Letters*.

WORK EXPERIENCE

Undergraduate Research Student, Dr. Bo Wang, University of Toronto/University Health Network

March 2024-
Present
(part-time)

Writing undergraduate thesis on genomic foundation model interpretability.

Developed methods for mechanistic interpretability of genomic foundation models. Implemented sparse autoencoders, automated feature description generation, and developed feature exploration and visualization for the Evo model.

Leading development and deployment of AI surgical scheduling system in hospital.

Implemented AI foundation model for accurate surgical duration prediction. Designed randomized control trial for evaluation (3-phase stepped wedge). First author of manuscript (in-progress) for *JAMA Cardiology*.

Undergraduate Research Student, Dr. Michael Hardisty, Sunnybrook

Sept 2024-Dec
2024 (part-time)

Implemented timeseries models to predict musculoskeletal degeneration in prostate cancer patients enabling personalized treatment.

Undergraduate Researcher, Dr. Angela Schoellig Learning Systems and Robotics Lab (LSY Lab), Technical University of Munich, Germany.

May-August 2024

[Preprint \(arXiv:2410.15185\)](#)- Developed AI-based robotic semantic safety filter. Used LLMs and VLMs to detect safety issues in a scene and generate robotic constraints.

Developed a [Lego-building robot](#) using Franka robot arm. Focused on AI-vision components including brick detection and pose tracking using foundation models.

Data Science Intern, Marsh, New York office.

June-August 2023

Developed a machine learning model for risk analysis in the healthcare field, the first of its kind in the field. Enables hundreds of hospitals to accurately assess their risk.

Research Student, SickKids Hospital with Dr. Sasha Litwin.

May-June 2023

Developed an AI chatbot designed to answer a parent's fever questions, improve at-home treatment, and prevent unnecessary ER visits.

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Research Student, Dr. Mike Fralick's Lab, Mount Sinai Hospital.

May 2022-May

Developed AI tools for healthcare. Full-time during summer, rehired to continue part-time during school year. Projects included:

2023

ChatBotMD

Created BERT-based AI chat bot prototype for clinical use, developed custom decision-making algorithms. Created chat bot website from scratch, set up networking, and run on AWS server with NGINX. Sole programmer.

Medical Literature Tool

Created website that uses my NLP and LLM approach to inform doctors of advances in medicine. Finds relevant papers and retrieves key information.

jrnowl.com

Using ML to automatically find journal publishing guidelines from medical journal websites for use on jrnowl.com

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

Academic Director & Academic Paper Read Group Mentor, University of Toronto Machine Intelligence Student Team (UTMIST).

2022-2023

Led monthly machine learning lectures. Topics taught included CNN, ConvNext, GNN, Variational Autoencoder, Attention, Transformer, etc.

Software Team Member, University of Toronto Design League Robodog Team.

2022

Responsible for inverse kinematics, and gait programming using ROS.

Competition Team Member, University of Toronto Concrete Toboggan Team

2021-2022

Satirical Newspaper Co-President, William Lyon Mackenzie Cl.

2020-2021

Math and Computer Science Program Peer Mentor & Calculus Tutor

2019-2021

Camp Counsellor, Green Acres Day Camp. Led activities for campers age 5-8

Summer 2019

TECHNICAL SKILLS

- Python, Julia, C, C++, Java, Matlab, Linux, ROS, Docker, HTML, CSS, SolidWorks
- LLMs, NLP, Computer Vision, PyTorch, JAX, Tensorflow
- University of Toronto Machine Shop Certification (2021)

AWARDS & ACHIEVEMENTS

Engineering Science Global Research Opportunities Program, Highly

2024

competitive scholarship among Engineering Science students. For summer research in Angela Schoellig's LSY Lab in Munich.

BioTalent Health and Biosciences Student Work Placement Wage Subsidy, For

2022

summer research in Dr. Mike Fralick's Lab.

Admission Scholarship, Faculty of Applied Science and Engineering, University of

2021

Toronto. \$2000

Advanced Placement (AP), 5/5 Computer Science A (2019), 5/5 Calculus AB(2021)

DECA Provincial Finalist & Case Award, Business Finance Series

2021

Honourable Mention, Queen's Model United Nations Conference

2021

3rd Place, LyonHacks Hackathon, William Lyon Mackenzie Collegiate Institute

2021

DECA Regional Top 60, Business Finance Series

2020